JaeHyuk Son

M.S. Candidate in Biomedical Signal Processing and Artificial Intelligence

Research interests include remote photoplethysmography (rPPG), blood pressure estimation, biomedical signal processing, deep learning, and digital healthcare.

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EDUCATION

M.S. in Electronic and Communication Engineering

Kwangwoon University   Aug. 2024 – Aug. 2026 (Expected)

- Thesis: “Physiological Distance-based Contrastive Forward-Forward Learning for Remote Blood Pressure Estimation Using Facial Videos”

B.S. in Electronic and Communication Engineering

Kwangwoon University   Mar. 2016 – Feb. 2023

PUBLICATIONS

Son, J. H. and Choi, Y. S. “Non-Contact Blood Pressure Estimation from Face Videos via Physiology-Aware Contrastive Learning.” Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026.

Jho, S. H., Kim, D. H., Son, J. H., and Choi, Y. S. “Physiology-Aware Multi-Branch Network for Fully Contactless Blood Pressure Estimation via Facial Remote Pulse Transit Time.” The International Joint Conference on Neural Networks (IJCNN), 2026.

Son, J. H. and Choi, Y. S. “Dual-Path Periodic Transformer for Enhancing rPPG Estimation.” 2025 International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC), 2025.

손재혁, 최영석. “얼굴 영상의 시공간 정보 활용 Transformer 기반 비접촉식 심박수 추정 기법” 대한전자공학회 학술대회, 2024.

손재혁, 최영석. “Mamba 기반 생리학적 특징 어텐션 프레임워크를 활용한 원격광용적맥파 기반 혈압 추정” 대한전자공학회 학술대회, 2026.

PATENTS

최영석, 손재혁. “얼굴 영상의 시공간 정보 활용 Transformer 기반 비접촉식 심박수 추정 시스템” Korean Patent Application No. 10-2024-0165318, 2024.

최영석, 손재혁. “생리학적 정보를 반영한 대조 학습 기반 얼굴영상에서의 비접촉 혈압 추정 방법” Korean Patent Application No. 10-2026-0006441, 2026.

HONORS & AWARDS

Outstanding Poster Paper Award
“노년층 발화 음성 다중모달 그래프 표현학습 기반 감정 인식”
The Korean Society of Medical & Biological Engineering (KOSOMBE) Spring Conference, 2024.

Award for “한국어 대화 기반 폭력 및 비폭력 상황 인식” using a BERT Language Model
Role: Korean conversational speech data collection
2021.

PROJECT EXPERIENCE

2021
“한국어 대화 기반 폭력 및 비폭력 상황 인식” (BERT-based Korean Dialogue Classification)

- **Responsible for collecting Korean conversational speech datasets.**
- **Received an award for the project.**

Research Intern, NeuroAI Laboratory, Kwangwoon University

- **Conducted research on speech and language AI.**